Detecting Depression from Interview Speech

Final model: voice + text + local LLM symptom scoring

Accuracy 79% (cross-validation), 78.8% on 33 unseen people

The task: screen for depression from one interview

Input: audio of a 7-20 minute clinical interview

Output: is PHQ-8 >= 10 (clinically depressed)? yes / no

Data: DAIC-WOZ + E-DAIC (USC), the standard benchmark

272

people interviewed

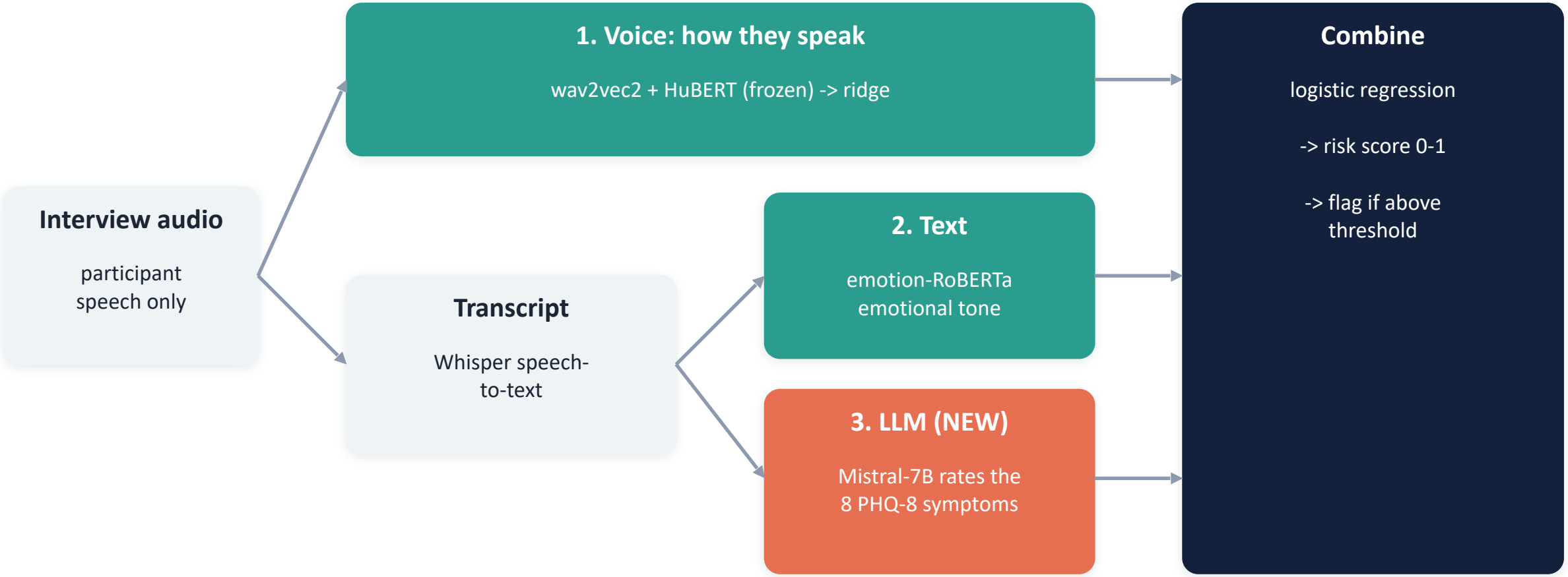
85

depressed (31%)

68.8%

accuracy of always guessing "not depressed" (the baseline)

How the final model works



Each part gives its own score; a small logistic regression combines the three into one risk score.

Every model in the pipeline

Step	Model	What it does	Trained by us?
Find speech	Silero VAD	Cuts out silence	No (pre-trained)
Voice	wav2vec2-base (layer 6), HuBERT-large-ER (layers 19, 20)	Turns speech into 768/1024 numbers describing the voice	No (frozen)
Speech-to-text	Whisper-tiny (OpenAI)	Writes down what the person said	No (pre-trained)
Text	emotion-RoBERTa-large	Turns the transcript into an emotion embedding	No (frozen)
Symptoms (NEW)	**Mistral-7B-Instruct v0.3, 4-bit, run locally**	Reads the transcript, rates 8 PHQ-8 symptoms 0-3	No (zero-shot)
Per-part learner	Ridge regression (x5)	Predicts the PHQ-8 score from each part	**Yes**
Combiner	Logistic regression	Merges the 3 scores into one risk score	**Yes**

Big models are used frozen; only small models are trained, which is right for 272 people (no overfitting).

The new idea: let an LLM read the interview

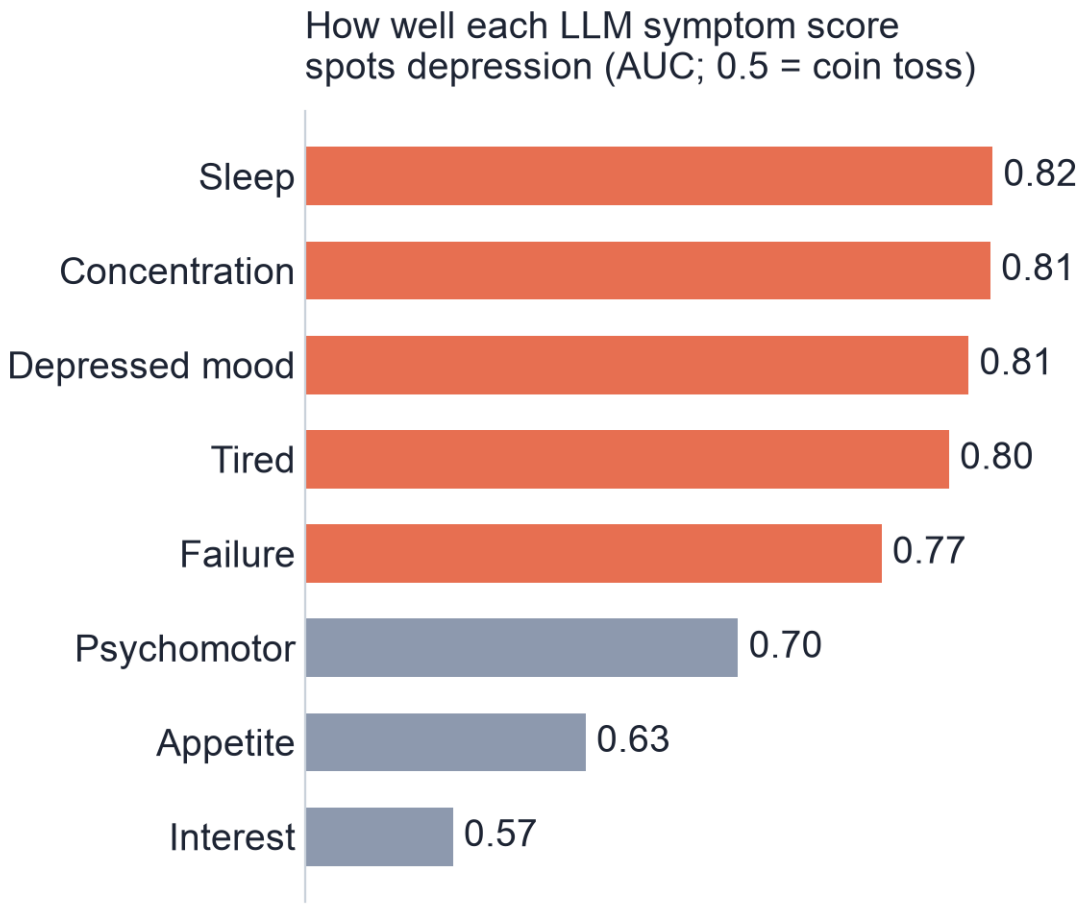
Prompt

"Estimate how the participant would answer each PHQ-8 item, 0-3."

Answer (participant 606)

```
interest 0.1, depressed 0.0, sleep
0.0, tired 0.4, appetite 0.0, failure
0.0, concentration 1.3, psychomotor
0.0
```

Never sees any label. No training. Runs offline on the laptop (Apple MLX).



Two honest tests, no data leakage

A Test A: cross-validation

All 272 people. Train on 4/5, test on the other 1/5, rotate until everyone is tested; repeat with 5 different splits.

Each person is predicted by a model that never saw them.

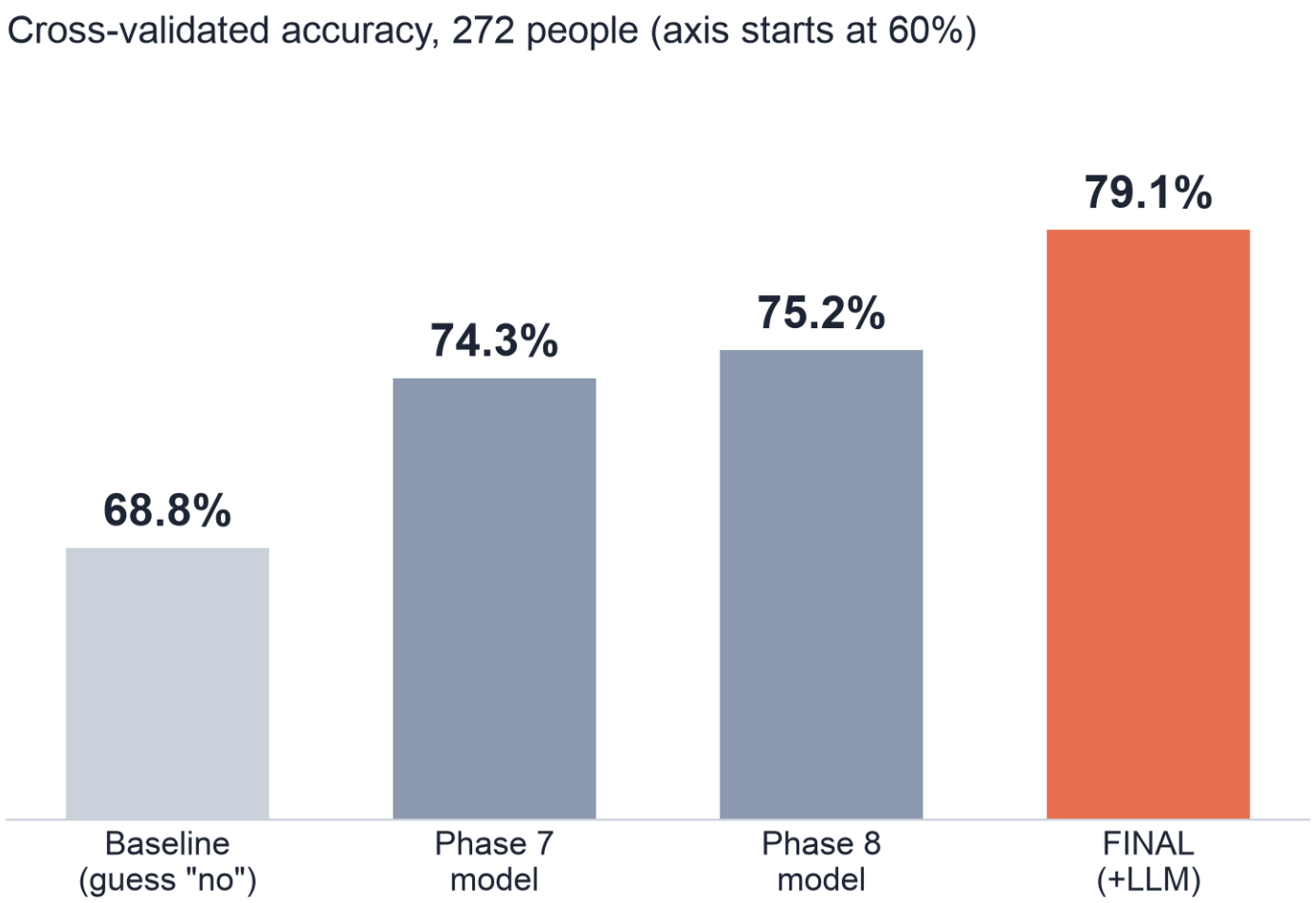
B Test B: unseen holdout

Train only on 239 people. Test once on 33 people kept aside in the "unseen" folder.

All settings (strengths, combiner, threshold) chosen on the 239 only.

Rules we kept: nobody appears in both training and test; the LLM never sees a label; interviewer questions removed from transcripts; plan and adoption rule written and pushed to GitHub before running.

Result A: accuracy rises from 74.3% to 79.1%



0.843

AUC (was 0.764)

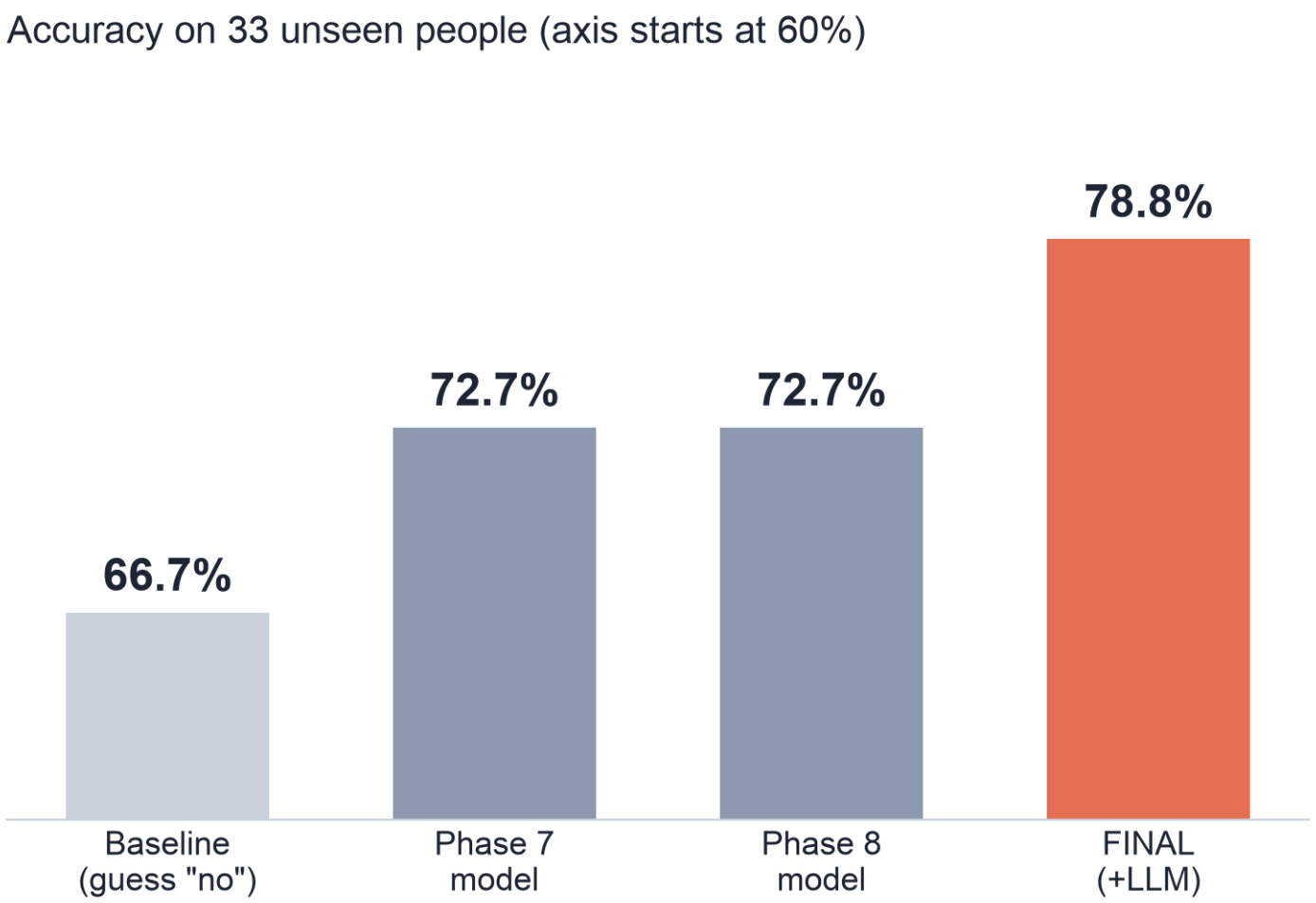
72.4%

balanced accuracy (was 67.0%)

77%

of depressed people caught at the balanced threshold (was 64%)

Result B: 26 of 33 unseen people correct



0.868

AUC (Phase 7: 0.777)

21 / 22

healthy people correctly cleared

9 / 11

depressed caught at the balanced threshold (Phase 7: 7)

Three models side by side

Metric	Phase 7 (74.3%)	Phase 8 (75.2%)	FINAL (+LLM)
Test A: cross-validation, 272 people (baseline 68.8%)			
Accuracy	74.3%	75.2%	**79.1%**
Balanced accuracy	67.0%	68.3%	**72.4%**
AUC	0.764	0.785	**0.843**
MCC / macro-F1	0.37 / 0.68	0.39 / 0.69	**0.49 / 0.74**
Test B: train 239, test 33 unseen (baseline 66.7%)			
Accuracy	72.7% (24/33)	72.7% (24/33)	**78.8% (26/33)**
AUC	0.777	0.785	**0.868**
Depressed caught (balanced thr.)	7 / 11	7 / 11	**9 / 11**

Final vs Phase 7: AUC +0.079 [95% CI +0.035, +0.124]. Final vs Phase 8: +0.057 [+0.015, +0.102]. Both significant.

Screenshot: the results as printed

Terminal: results

```
$ python run_project.py --report

Phase 9: nested cross-validation, N = 272 (baseline 68.8%)
   model                              AUC      acc  bal-acc   sens   spec
   P4  (74.3% model)                  0.764    74.3%    67.0%    48%    87%
   R0_voice_text  (75% model)         0.785    75.2%    68.3%    50%    87%
   F1_voice_text_llm  (FINAL)         0.843    79.1%    72.4%    55%    90%

Holdout: train 239 -> test the 33 unseen (baseline 66.7%)
   model                              AUC      acc  bal-acc   sens   spec
   P4  (74.3% model)                  0.777    72.7%    68.2%    55%    82%
   R0_voice_text  (75% model)         0.785    72.7%    70.5%    64%    77%
   F1_voice_text_llm  (FINAL)         0.868    78.8%    70.5%    45%    95%

Final saved model: models/final_phase9_f1.joblib  (trained on all 272)
```

Top block: Test A, cross-validation on 272.

Bottom block: Test B, trained on 239, tested on 33 unseen.

Orange = final model.

acc = accuracy, bal-acc = balanced accuracy, sens = depressed caught, spec = healthy cleared.

Screenshot: the app on one recording

Terminal: predict_audio.py on one recording

```
$ python predict_audio.py NEW_AUDIO_DAIC/606_AUDIO_interviewee.wav --domain avatar

Scoring PHQ-8 items with the local LLM ...
LLM PHQ-8 item estimates (0-3): interest 0.1, depressed 0.0, sleep 0.0,
   tired 0.4, appetite 0.0, failure 0.0, concentration 1.3, psychomotor 0.0 | sum 1.9

Depression risk score: 0.036  (0 = low, 1 = high)
  Max-accuracy threshold 0.542: not flagged
  Balanced threshold     0.196: not flagged

Model: Phase 9 F1, nested-CV AUC 0.84 / accuracy 79%; unseen-33 holdout 78.8%.
(True PHQ-8 of participant 606 = 5: not depressed -> correct)
```

One command: audio in, risk score out.

Shows the 8 LLM symptom ratings, so the result is explainable.

Output matches the training pipeline exactly (score 0.036).

Runs fully offline on an 8 GB MacBook in about 1-3 minutes. Screening aid only, not a diagnosis.

Why we trust it, and its limits

Why it is real

- Both tests agree: 79.1% and 78.8%
- Without the 17 people who say "I am depressed": AUC still 0.835
- Works for both interview setups (AUC 0.84 / 0.86) and both genders
- LLM never sees labels; nothing tuned on test people

Limits

- 33 unseen people is small: 1 person = 3 points
- The 33 were opened once before (Oct 19), so a repeat check
- At the max-accuracy threshold only about half of depressed people are caught; use the balanced threshold for screening
- Research prototype, not a diagnosis

79% accuracy, AUC 0.84

Adding local LLM symptom scoring to voice + text raised accuracy from 74.3% to 79.1% (cross-validation) and from 72.7% to 78.8% on 33 unseen people, against a 68.8% / 66.7% baseline.